

Evolution Strategies as Infinite-Dimensional Backpropagation: A Non-Standard Analysis Perspective on Biological Evolution

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Abstract

Recent advances in hyperscale evolution strategies (ES), such as the "Eggroll" architecture, have demonstrated that gradient-free optimization can scale comparably to backpropagation-based methods. This paper proposes a theoretical framework interpreting ES not merely as a zeroth-order approximation, but as an infinite-dimensional backpropagation process via the lens of non-standard analysis. We demonstrate that the aggregation of infinite random perturbations (mutations) in ES is mathematically equivalent to computing the gradient of a Gaussian-smoothed loss landscape. Furthermore, we argue that this equivalence provides a mathematical foundation for understanding biological evolution: living organisms function as discrete samples in a massive parallel computation, collectively approximating the gradient of the survival function. This suggests that biological evolution is effectively performing backpropagation at a planetary scale without explicit differentiation.

1 Introduction

Deep learning has traditionally relied on Backpropagation (BP) to compute gradients $\nabla L(\theta)$ for parameter updates. In contrast, Evolution Strategies (ES) update parameters by sampling random perturbations in the parameter space. While ES has been viewed as a scalable alternative to Reinforcement Learning, the precise mathematical relationship between the "blind" search of ES and the "directed" descent of BP has often been simplified as a mere finite difference approximation.

In this paper, we revisit this relationship using concepts from non-standard analysis. We show

that as the population size $N \rightarrow \infty$, the accumulation of ES updates converges to the exact gradient of a smoothed objective function. This implies that "integration" (ES) and "differentiation" (BP) are unified at the limit. We further discuss the philosophical implications of this duality, proposing that biological evolution can be interpreted as an implicit backpropagation process performed by the biosphere.

2 Mathematical Equivalence

2.1 Evolution Strategies (ES)

Let θ be the parameters of a neural network (or an organism's DNA), and $L(\theta)$ be the loss (or negative fitness) function. ES estimates the descent direction by applying Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$. The update rule for infinite population size is defined as the expectation:

$$\nabla_{\text{ES}} = \frac{1}{\sigma} \mathbb{E}_{\epsilon}[L(\theta + \sigma\epsilon) \cdot \epsilon] \quad (1)$$

This equation represents a search based purely on the correlation between the perturbation ϵ and the resulting loss L , without calculating any derivatives.

2.2 Smoothed Backpropagation

Consider the "smoothed" loss function $L_{\sigma}(\theta)$, defined as the convolution of the original loss landscape with a Gaussian kernel. The gradient of this smoothed landscape is:

$$\nabla L_{\sigma}(\theta) = \nabla \mathbb{E}_{\epsilon}[L(\theta + \sigma\epsilon)] = \mathbb{E}_{\epsilon}[\nabla L(\theta + \sigma\epsilon)] \quad (2)$$

The right-hand side represents the average of gradients calculated by backpropagation at infinite points in the neighborhood of θ .

2.3 Stein’s Lemma and Unification

Using Stein’s Lemma (Gaussian integration by parts), we can prove the fundamental identity linking these two approaches:

$$\underbrace{\frac{1}{\sigma} \mathbb{E}_{\epsilon}[L(\theta + \sigma\epsilon) \cdot \epsilon]}_{\text{ES (No Derivative)}} = \underbrace{\mathbb{E}_{\epsilon}[\nabla L(\theta + \sigma\epsilon)]}_{\text{Average BP (Infinite Derivatives)}} \quad (3)$$

This equality is non-trivial. It states that the weighted average of zero-order evaluations (ES) is mathematically identical to the expected value of first-order gradients (BP).

3 Interpretation via Non-Standard Analysis

Non-standard analysis allows us to rigorously treat infinitesimals and infinite numbers.

- **BP as Local/Microscopic:** Standard backpropagation computes the derivative at a single point θ , strictly relying on local differentiability.
- **ES as Global/Macroscopic:** ES integrates over a hyper-volume defined by σ .

We propose that ES can be viewed as "Backpropagation in the limit of infinite sampling." In the context of non-standard analysis, the integral operation in ES acts as a "global derivative." The "monad" (infinitesimal neighborhood) around θ is explored by the population. The "consensus" of this infinite population perfectly recovers the gradient information.

4 Biological Implications: Life as Calculation

This mathematical framework offers a profound perspective on the nature of life.

4.1 The Biosphere as a GPU

If we consider the biosphere as a computational system:

- Each organism is a sample $\theta + \sigma\epsilon_i$.
- The life and death of the organism corresponds to the evaluation of $L(\theta + \sigma\epsilon_i)$.

- Reproduction corresponds to the parameter update $\theta_{t+1} \leftarrow \theta_t - \eta \nabla_{\text{ES}}$.

4.2 The Purpose of Existence

Under this interpretation, the "purpose" of an individual organism’s existence is to serve as a probe for gradient estimation. Nature cannot perform analytical differentiation (it does not know the equation of physics), but it can perform integration through massive parallel sampling. By summing up the lives of billions of individuals, the species effectively computes the gradient ∇L , moving towards better adaptation. Thus, every life is a necessary computational unit to drive the "infinite backpropagation" of the species.

5 Conclusion

We have demonstrated that Evolution Strategies are not merely an approximation of backpropagation but its dual representation through the lens of infinite sampling. This duality suggests that computational intelligence and biological evolution share the same underlying mathematical principle: the extraction of gradient information from high-dimensional noise.

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References

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